MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732377
MIRROR DRIVE ELECTRONICS

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732377 Mirror Drive Electronics which is the A10 component of the M21 Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-45662	Calibration System Requirements
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-STD-45662	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732377	Circuit Card Assembly: 11732377 Mirror Drive Electronics
11732629	Amplifier, Operational
11743670	Motor - Generator Gear Head

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750427	Adapter, Computer Board - A10
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
11821500	Vibration/Shock - Test Fixture - Computer Unit FAT
11821520	Automatic Test Assembly, Central Processor Unit
11821854	Test Fixture Auto Card, Test Set, Computer, FAT

Packaging Data Sheet

P11732377	Packaging of Circuit Card Assembly, Mirror Drive Electronics: 11732377

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 *Fabrication*. The mirror drive electronics circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732377 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Signals. The assembly shall provide the following signals:

Signals designation	Signal Nomenclature
a. D_L	Mirror deflection
b. MDM1 and MDM2	Mirror motor drivers
c. MDE O.K.	Mirror drive electronics fail flag
d. RFOK	Rangefinder fail flag
e. $ST_{(L)}$	Stationary tank bar

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{o} \pm 18^{o}F$
b.	Relative humidity	50 percent + 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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3.3.1 D_L output. With the power and signals (table I) applied, and the inputs (table II) applied, the nominal output of P1-4B shall be described by the following transfer function:

$$V_{4B} = - V_{2A} - 0.03123 * V_{1A} * SW1 - (0.00293 * SW1 + 0.00710) * V_{2B}$$

TABLE I. Loads, power, signal and logical sources.

Item	Conditions	Characteristics	Connections
1.0	Load		
1.1	Mirror Drive Load Circuit	See figure 1	Connected between P1-22B and P1-19A
2.0	Power		Between pins:
2.1	24.0 vdc (24V COMP 3)	Tolerance: ± 6.0 vdc Ripple: 500 mv p-p max	P1-25A and P1-23A (rtn)
2.2	15.000 vdc (+15V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-6A and P1-3A (rtn)
2.3	-15.000 vdc (-15V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-21A and P1-3A (rtn)
2.4	5.000 vdc (+5V)	Tolerance: ± 0.500 vdc Ripple: 100 mv p-p max	P1-20A and P1-13A (rtn)
3.0	Signal Sources	As specified below 1/	Applied between the following pins and P1-13A (rtn):
3.1	AC REF 1	Sine wave, phase 1, 5.000 ± 0.010 vrms	P1-2B
3.2	MR1	See figure 2	P1-5A
3.3	MR2	See figure 2	P1-22A
4.0	Logical Sources	As specified below 2/	Applied between the following pins and P1-13A (rtn):
4.1	BS	(Logical 1) 15.0 ± 1.0 vdc (Logical 0) -15.0 ± 1.0 vdc	P1-9A

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TABLE I. Loads, power, signal and logical sources. - Continued

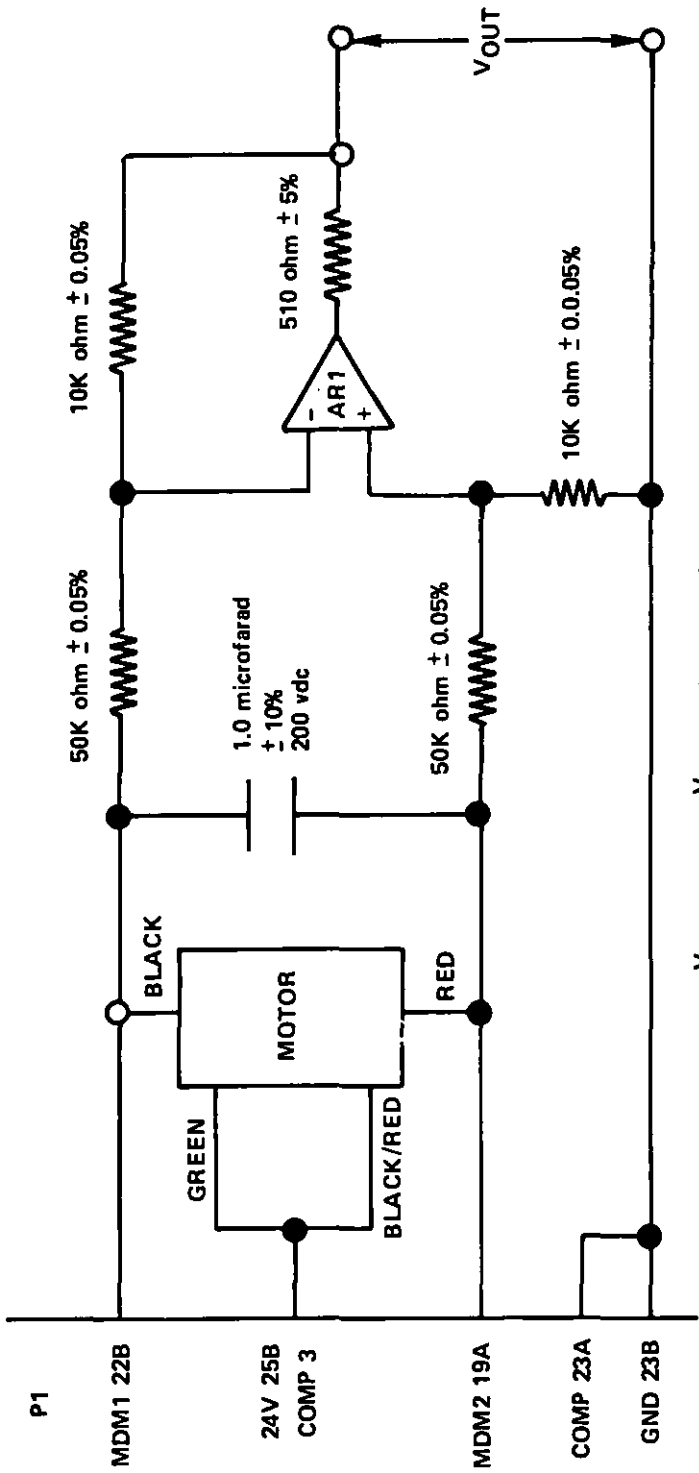
Item	Conditions	Characteristics	Connection
4.2	AUTO RANGE SELECT	(Logical 1) 3.5 + 1.5 vdc (Logical 0) 0.5 ∓ 0.5 vdc	P1-11A
4.3	BR(L)	Same characteristics as item 4.2	P1-10A
4.4	ST BAR	Same characteristics as item 4.2	P1-18A
4.5	ROK	Same characteristics as item 4.2	P1-15B
5.0	Inter-connections		Connected between following pins:
5.1		Jumper wire	P1-3A and P1-13A
5.2		Jumper wire	P1-13B and P1-23B
5.3		Jumper wire	P1-3B and P1-23A

1/ a. Positive (+) magnitudes (phase 1 signals) shall be in phase ± 1.0° with item 3.2.

b. Negative (-) magnitudes (phase 2 signals) shall be 180° ± 1.0° out of phase with item 3.2.

2/ Logical terms have a value of either one (1) or zero (0) as defined by the voltage levels in the characteristics column.

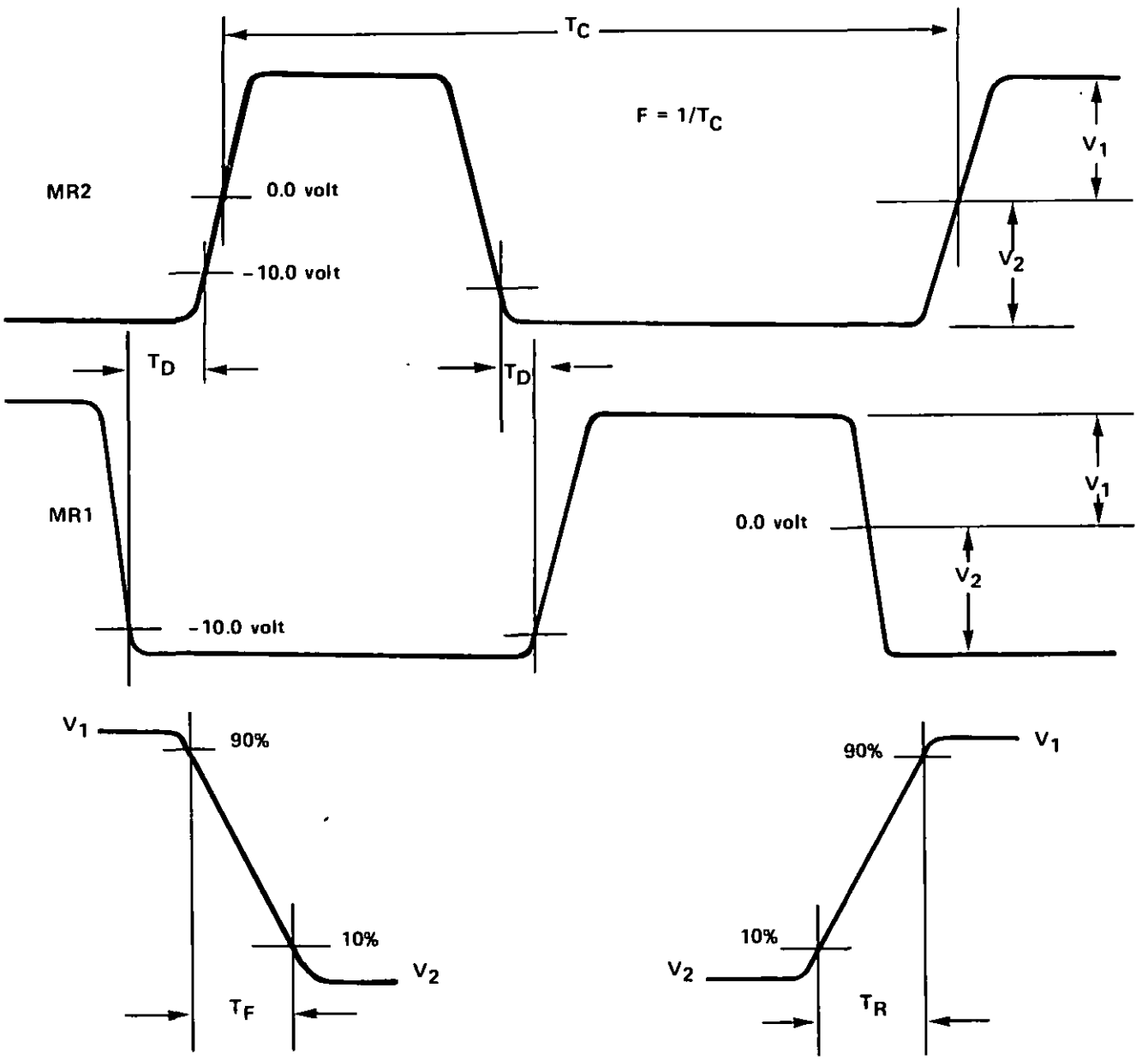
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a. $V_{MOTOR} = 5 * V_{OUT}$ (SEE 3.3.2).
b. SEE DRAWING 11743670 FOR MOTOR REQUIREMENTS.
c. SEE DRAWING 11732629 FOR AR1 REQUIREMENTS.

FIGURE 1. Mirror drive load circuit.

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	Standard ambient	140° F	-25° F
V_1	15 ± 0.50 volts	15 ± 0.50 volts	15 ± 0.50 volts
V_2	-15 ± 0.50 volts	-15 ± 0.50 volts	-15 ± 0.50 volts
T_D	1 to 30 microsecond	1 to 30 microsecond	1 to 35 microsecond
T_R	0.5 to 5 microsecond	0.5 to 4 microsecond	1 to 6 microsecond
T_F	0.5 to 4 microsecond	0.5 to 4 microsecond	0.5 to 6 microsecond
F	400 ± 1 Hz	402 ± 2 Hz	398 ± 2 Hz

FIGURE 2 Master reference waveforms.

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TABLE II. D_L inputs.

Item	Input Signal	Pin Location 1/	Signal Limits
1	V_{2A}	P1-2A	0 to 5 vrms, phase 1 (+) or phase 2 (-)
2	V_{1A}	P1-1A	0 to 5 vrms, phase 2 (-)

1/ Applied with respect to P1-3A.

where term SW1 has the value of either 1 or 0 as defined by the logical expression:

$$SW1 = BS \text{ or } \overline{\text{AUTO RANGE SELECT}} \text{ or } \overline{BR(L)}$$

The tolerance of the output shall not be greater than the value of the following equation:

$$\Delta V_{4B} = 0.0005 * \left|V_{4B}\right| + 0.0005 * \left|V_{2A}\right| + 0.0001 * SW1 * \left|V_{1A}\right| + (0.0001 * SW1 + 0.0001) * \left|V_{2B}\right| + 0.001$$

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$\left|V_{4B}\right| < 7$ volts root mean square (vrms) (see 4.6.2.1)

3.3.2 MDM1 and MDM2 outputs. With the load (item 1.1 of table I) connected, power and signals (table I) applied, and the inputs (table III) applied, the nominal motor drive voltage, see figure 1, shall be described by the following transfer function:

$$V_{MOTOR} = 9.88 * SIG2$$

where:

a. $SIG2 = -1.00 * SIG1 + 1.00 * V_{4A}$

b. $SIG1 = -1.112 * SIG3$

c. $SIG3 = -1.00 * SIG4$

d. $SIG4 = -0.908 * SIG5$

e. $SIG5 = -20.0 * (V_{4B} - 1.6978 * V_{5B})$

f. V_{4B} is defined in 3.3.1.

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TABLE III. MDM1 and MDM2 inputs.

Item	Input Signal	Pin Location 1/	Signal Limits
1	V_{4A}	P1-4A	0 to 5 vrms, phase 1 (+) or phase 2 (-)
2	V_{4B}	P1-4B	See 3.3.1
3	V_{5B}	P1-5B	0 to 5 vrms, phase 1 (+) or phase 2(-)

1/ Applied with respect to P1-3A.

The tolerance of the output shall not be greater than the value of the following equation:

$$\Delta V_{MOTOR} = 0.61 * |SIG2| + 9.88 * \Delta SIG2 + 0.1$$

Where:

a. $\Delta SIG2 = 0.011 * |SIG2| + 0.001 * |SIG1| + 0.021 * |V_{4A}| + 1.00 * \Delta SIG1$

b. $\Delta SIG1 = 0.010 * |SIG3| + 1.112 * \Delta V_{SIG3} + 0.004$

c. $\Delta SIG3 = 0.020 * |SIG4| + \Delta SIG4 + 0.005$

d. $\Delta SIG4 = 0.010 * |SIG5| + 0.908 * \Delta SIG5 + 0.006$

e. $\Delta SIG5 = 0.54 * V_{4B} - 0.92 * V_{5B} + 20.0 * V_{4B} + 20.0 * \Delta V_{4B}$

f. V_{4B} is defined in 3.3.1.

Application of the transfer function and the tolerance equation shall be valid within the following limits:

a. $|SIG5| < 5$ vrms

b. $|V_{MOTOR}| < 15$ vrms (See 4.6.2.2)

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3.3.3 MDE O.K. output. With the load (item 1.1 of table I) connected, power sources of table I applied, signal sources (items 3.2, 3.3 and 3.4 of table I) applied, pins P1-2A, P1-2B, P1-5B and P1-10A connectd to pin P1-13A and the AUTO RANGE SELECT logical source (item 4.2 of table I) set to Logical 1, the output voltage at pin P1-15A shall perform as specified in 3.3.3.1 through 3.3.3.5.

3.3.3.1 Malfunction state transition. The voltage at pin P1-15A shall change from 4.5 + 1.0 vdc to 0.5 + 0.5 vdc with the input at pin P1-4A varied from 0.00 ± 0.01 vrms to 1.35 ± 0.01 vrms phase 1.

3.3.3.2 Malfunction state mirror drive voltage. With the occurence of the transition of 3.3.3.1, V_{MOTOR} shall be not less than 10.00 vrms.

3.3.3.3 Nominal indication transition. The voltage at pin P1-15A shall change from 0.5 + 0.5 vdc to 4.5 + 1.0 vdc with the input at pin P1-4A varied from 1.35 ± 0.01 vrms to 0.00 ± 0.01 vrms, phase 1.

3.3.3.4 Normal mirror drive voltage. With the occurrence of the transition of 3.3.3.3, V_{MOTOR} shall be not greater than 14.00 vrms.

3.3.3.5 Phase 2 excitation. The requirements of 3.3.3.1 through 3.3.3.4 shall be met with pin P1-4A excited with phase 2 signals.

3.3.4 Self test logic. With the load (item 1.1 of table I) connected, power sources of table I applied, signal sources of table I applied, input signals (item 1 of table II and item 1 of table III) applied, pin P1-15A connected to pin P1-18A and pins P1-1A, P1-2B, P1-5B, P1-9A and P1-10A connected to pin P1-13A, the outputs of table IV shall be generated with the corresponding inputs established for each of the 8 items. (See 4.6.2.4)

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TABLE IV. Self test logic.

Item	Inputs 1/					Outputs 2/		
	V_{2A}	AUTO RANGE SELECT	ROK	P1-25B	V_{4A}	P1-12B	P1-15A	P1-17A
1	0.0	1	0	OFF	0.0	0	1	0
2	0.0	1	1	OFF	0.0	1	1	0
3	0.3	1	1	OFF	0.0	0	0	1
4	0.3	0	0	OFF	0.0	0	1	0
5	0.0	0	1	ON	1.5	0	1	0
6	0.0	1	1	ON	1.5	0	0	1
7	0.3	1	0	ON	0.0	0	1	0
8	0.3	0	1	ON	0.0	0	1	0

1/ a. See table I for logic state definitions.
b. 0.00 may be an interconnection with P1-3A.
c. 0.30 = 0.30 ± 0.02 vrms phase 1.
d. OFF = 0.0 ± 0.1 vdc.
e. ON = Interconnection with P1-25A.
f. 1.5 = 1.5 ± 0.1 vrms phase 1 or phase 2.

2/ a. 1 = 4.5 ± 1.0 vdc measured with respect to P1-3A.
b. 0 = 0.5 ± 0.5 vdc measured with respect to P1-3A.

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3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 3. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4, 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

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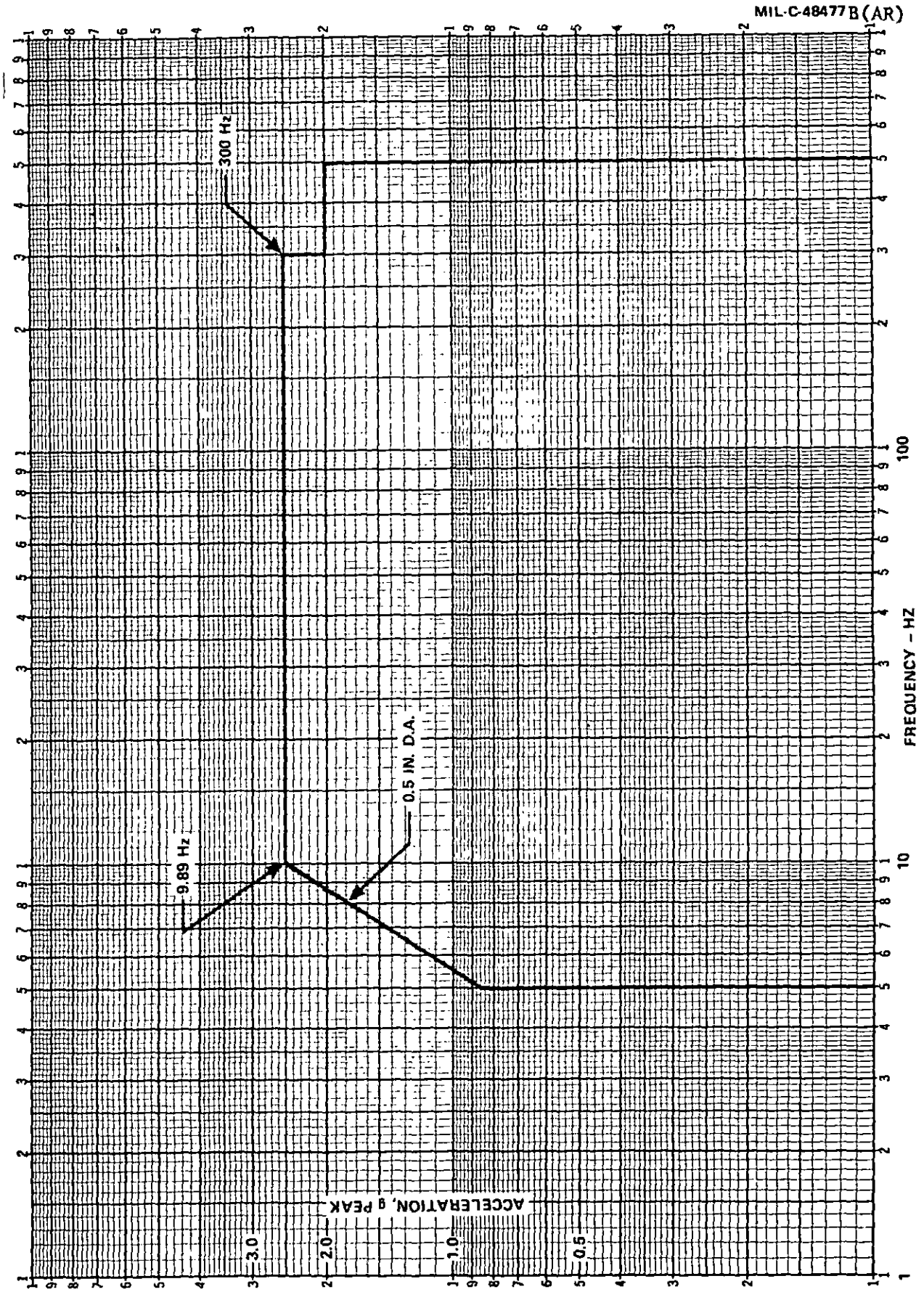


FIGURE 3. Vibration profile.

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4.4 First article. The requirements for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables V, VI and VII.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly.	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VI.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VII shall be applied separately to each assembly in the sample.

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4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet the specified requirements, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VI and only the test of item 307 in table VII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table V and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732377.

TABLE V. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101	D_L output	3.3.1	4.6.2.1
102	MDM1 and MDM2 outputs	3.3.2	4.6.2.2
103	MDE O.K. comparator output	3.3.3	4.6.2.3
104	Self test logic	3.3.4	4.6.2.4

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TABLE VI. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

TABLE VII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature Cycling	3.4.3 and 3.4.4	4.6.3.5

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table VIII shall be used where specified to perform the required tests.

4.6.1.1 AC voltage measurement. Unless otherwise specified, all ac voltage measurements refer to the in-phase component of the fundamental where MR1 (table I, item 3.3) is the phase reference.

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TABLE VIII. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750427	Adapter, Computer Board - A10
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock-Z
5	11821500	Vibration/Shock - Test Fixture - Computer Unit FAT
6	11821520	Automatic Test Assembly, Central Processor Unit
7	11821854	Test Fixture Auto Card, Test Set, Computer, FAT

4.6.2 Performance tests.

4.6.2.1 D_I output. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.1

4.6.2.2 MDM1 and MDM2 outputs. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.2

4.6.2.3 MDE O.K. output. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.3

4.6.2.4 Self test logic. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.4

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table VIII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 $\pm$ 4 g with a time duration of 18 $\pm$ 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

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4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table VIII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 +10.0 g with a time duration of 1.5 +0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 5 of table VIII or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 3 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table VIII or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table V shall be applied.

4.6.3.4 Low temperature. Use item 7 of table VIII or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table V shall be applied.

4.6.3.5 Temperatue cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 4. At the conclusion of this test, subject the assembly to the performance tests of table V.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732377. The level of protection shall be as specified in the procurement document. (See 4.5.6)

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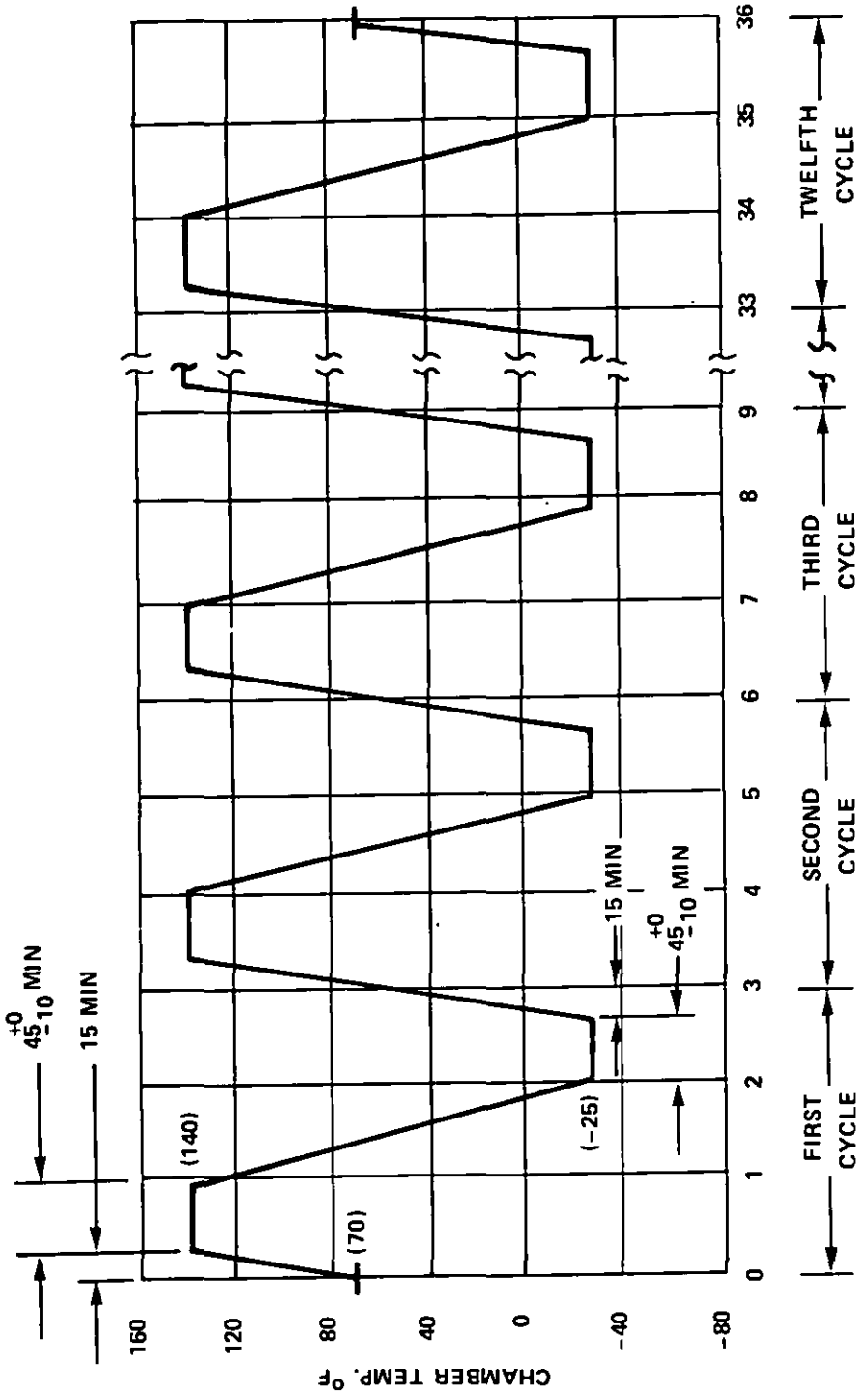


FIGURE 4. Temperature cycling.

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6. NOTES

6.1 Intended use. The Mirror Drive Electronics circuit card assembly is used in the M21 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Special tests and examinations as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48477B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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MIL-C-48477B CIRCUIT CARD ASSEMBLY: 11732377 MIRROE DRIVE ELECTRONICS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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